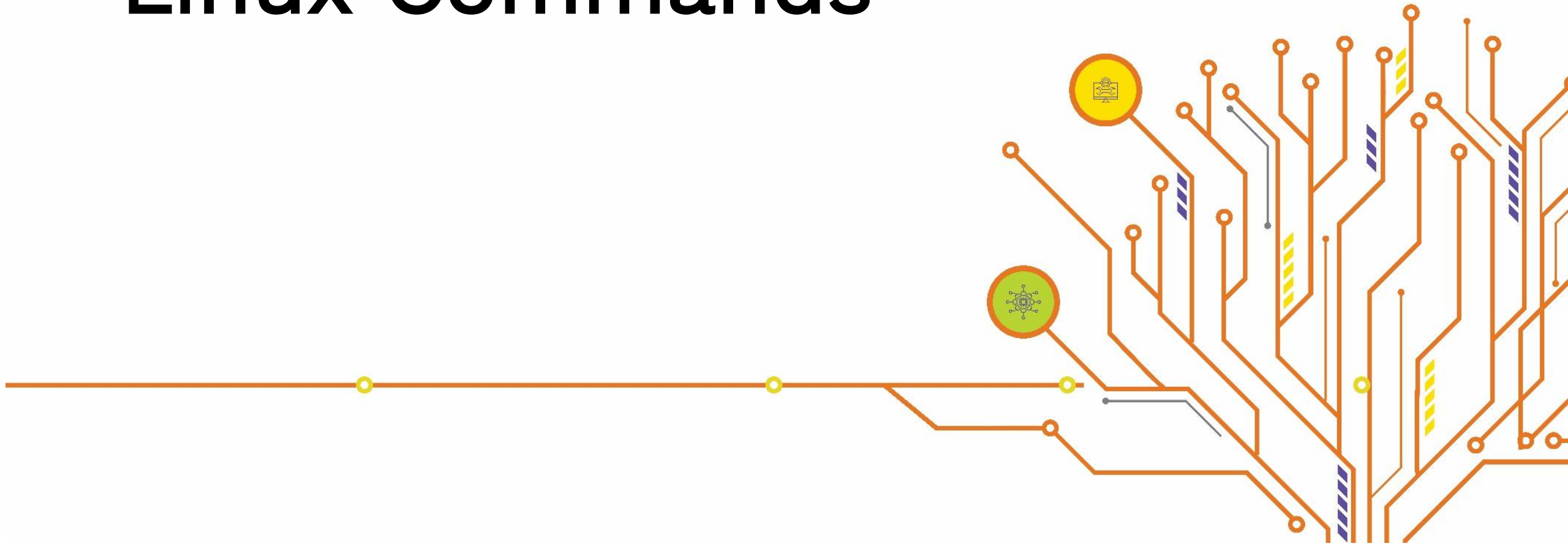
KPIT | APEX LAB

Linux Commands



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Directory Navigation and File Operations

- **pwd** # Show current directory
- **cd /path/to/dir** # Change directory
- **ls -l** # List files with details
- **mkdir new_folder** # Create a directory
- **rm filename.txt** # Remove a file
- **rmdir old_folder** # Remove an empty directory
- **cp file1 file2** # Copy file
- **mv file1 newname** # Rename or move file

Example

```
$ mkdir project                    # create directory - project
$ cd project                       # go to project directory
$ touch main.c                     # creae main.c file
$ ls –l                            # list all the files with details
$ cp main.c backup_main.c          # copy main → backup_main
$ mv backup_main.c old_main.c      # move backup_main → old_main
```

Task 1 - Exploring

1. Show your current directory.
2. Move to /tmp and list all files (including hidden ones).
3. Go back to your home directory.

Task 2 - Creating & Removing

- Create a directory named “project_logs”
- Inside it, create an empty file named “debug.txt”
- Copy debug.txt to debug_backup.txt.
- Rename debug.txt → debug_main.txt.
- Delete debug_backup.txt.
- Try removing the folder — observe error.
- Now delete the file inside and remove the directory.

Task 3 - Creating & Removing (1/3)

Your team lead says:

- “We’re starting Project ORION.
- I need you to create a clean directory structure, generate some placeholder files, organize them, make backups, move some logs, archive a vendor drop, and search through files to extract important information.”
- Tasks you must complete on next slide.

Directory	Files to Create (with contents)
orion_simple/	*(root folder)*
docs/	**design_A.md** – 3–5 lines describing "Design A" (module, rate, interface). **design_B.md** – 3–5 lines describing "Design B" (module, protocol, constraints). **requirements.txt** – 5–8 bullets: core features & constraints for ORION.
src/	**main.c** – a tiny C program that prints Hello ORION\n once. **preprocess.py** – prints Data pipeline started.
data/	*(no files directly here)*
data/raw/	**measure.csv** – header ts,value + ~10 timestamped rows (e.g., 2026-02-16T12:00:00Z,42).
data/processed/	*(empty initially; later must contain a **copy** of raw/ as import_Q1/)*
logs/	**app.log** – 30+ lines like INFO app: step <n> (n=1..30). **sys.log** – 20+ lines like WARN sys: event <n>.
config/	**dev.env** – DB_HOST=localhost, DB_PORT=5432, ENV=dev (one per line). **prod.env** – DB_HOST=10.0.0.5, DB_PORT=5432, ENV=prod.
vendor/	*(no files directly here)*
vendor/drop_Q1/	*(this folder will be **moved** later into data/archive/)*
vendor/drop_Q1/firmware/	**fw_readme.txt** – 2–3 lines: firmware version, date, short note.
vendor/drop_Q1/notes/	**release_notes.txt** – 3–5 bullets: major changes & known issues.

Task 3 - Creating & Removing (3/3)

After creating the base tree and files, students must also:

- Copy docs/design_A.md → docs/ORION_rfp_v1.md
- Copy data/raw/ → data/processed/import_Q1/ (recursive)
- Move vendor/drop_Q1/ → data/archive/vendor_drop_Q1/
- Hard-link docs/design_B_link.md → docs/design_B.md (same inode)
- Symlink docs/app_log_symlink → ../logs/app.log
- Do safe (rm -i) vs recursive (rm -r) deletes
- Search with grep/find (by content, size, time), and verify with ls -l, ls -i, and (if available) tree

Permissions and Access Control

- ls –l # list the files with all permissions
- chmod 660 file # change file permissions, in order: user group others, 6 = rw-, 0 = ---
- chown user:group /dev/mydevice

Task 4 - Inspect

- Run:

  ls -l /dev/shm

- Identify:
  - Owner
  - Group
  - Permission bits

Task 5 - Modify Permissions

- Create a dummy file

  echo "sample" > testfile
- Change the permissions to read-write for owner & group only
- Check the file permissions again.

Task 6 - Ownership

- Create a new group “students”
- Change group ownership of the file

Process and Thread Management

- **ps -ef | grep lane_detect** → find running process
- **pidof ./a.out** → get PID directly
- **strace -f -e trace=process ./lane_detect** → trace forks/execs
- **kill -9 <pid>** → forceful termination
- **nice -n 10 ./a.out**
- **renice -n -5 <pid>**
- **pstree -p <pid>**

Task 7 - Identify Processes & Trace activity

1. Start a sample program or sleep process
   sleep 500 &
2. Find its PID using:
   - ps -ef | grep sleep
   - pidof sleep
3. Trace Its Activity & observe which all system calls are logged

   strace -f -e trace=process sleep 1

Inode Inspection and File Metadata

- **ls –i** → shows inode count; important for debugging “No space left on device” even when disk space is available.
- **stat filename.txt**
- **find . -inum <inode>**
- **df -i**

Monitoring CPU and Memory Usage

- top
- vmstat 1
- free -h